Part IIB Revision Notes

4M24: Computational Statistics and Machine Learning

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1 Monte Carlo Methods

Central limit theorem for Monte Carlo: The Monte Carlo estimate follows

$$\frac{1}{N}\sum_n f(x_n) \sim \mathcal{N}\left(\mathbb{E}[f(x)], \frac{\sigma_f^2}{N}\right) \text{ as } N \to \infty$$

where σ_f^2 is the variance of $f(x)$.

Importance sampling: use another probability density $q(x)$ that satisfy $q(x) \neq 0$ when $f(x) \neq 0$ to sample x. The expectation can be rewritten as

$$\begin{aligned}\mathbb{E}_p[f(x)] &= \int f(x)p(x)dx \\ &= \int f(x)\frac{p(x)}{q(x)}q(x)dx \\ &= \mathbb{E}_q\left[f(x)\frac{p(x)}{q(x)}\right]\end{aligned}$$

The orginal Monte Carlo estimate $\frac{1}{N}\sum_n f(x_n)$ with each $x_n \sim p(x)$ becomes $\frac{1}{N}\sum_n f(x_n)\frac{p(x_n)}{q(x_n)}$ with each $x_n \sim q(x)$. The difference in variances is

$$\sigma_f^2 - \sigma_{\tilde{f}}^2 = \int f(x)^2 p(x)dx - \int f(x)^2 \frac{p(x)^2}{q(x)}dx$$

The biggest decrease in variance is achived when $q(x) \propto |f(x)|p(x)$.

Control variates: use another function $g(x)$ with known expectation $\mathbb{E}[g(x)]$ to reduce the variance of the Monte Carlo estimate. The new estimator is

$$\hat{E}_f^c = \hat{E}_f + c(\hat{E}_g - \mathbb{E}[g(x)])$$

where $\hat{E}_f$ and $\hat{E}_g$ are the Monte Carlo estimates of $f(x)$ and $g(x)$, respectively.

The optimal value of c is

$$c^* = -\frac{\text{Cov}(\hat{E}_f, \hat{E}_g)}{\text{Var}(\hat{E}_g)}$$

For multiple control variates, the estimator is $\hat{E}_f^c = \hat{E}_f + \sum_i c_i(\hat{E}_{g_i} - \mathbb{E}[g_i(x)])$, and the optimal coefficients are $\mathbf{c}^* = -C^{-1}\mathbf{b}$, where C is the covariance matrix of $\hat{E}_{g_i}$ and $\mathbf{b}$ is the covariance vector between $\hat{E}_f$ and $\hat{E}_{g_i}$.

2 Lebesgue Integral and Measure

Riemann integral: Define two summations for a function f on an interval $[a, b]$ and $\Delta x_k = [x_k, x_{k+1}]$:

$$S_p(x) = \sum_{k=1}^{n} M_k(x_{k+1} - x_k) \text{ with } M_k = \sup_{x \in \Delta x_k} f(x)$$
$$s_p(x) = \sum_{k=1}^{n} m_k(x_{k+1} - x_k) \text{ with } m_k = \inf_{x \in \Delta x_k} f(x)$$

Define the limit of the summations as $n \to \infty$: $S(f) = \liminf_{n\to\infty} S_p(x)$, $s(f) = \limsup_{n\to\infty} s_p(x)$. If $S(f) = s(f) = A$, then A is the Riemann integral $\int_a^b f(x)dx$.

Limitations:

- The function must be continuous almost everywhere.
- The construction of the integral requires a parition of the domain.
- The exchange of limit and integral is not always valid.

Convergence of functions:

- Pointwise convergence: Given $\epsilon > 0$, there exists a natural number $N(\epsilon, x)$ such that for $n > N$, $|f_n(x) - f(x)| < \epsilon$. Does not preserve continuity.
- Uniform convergence: Given $\epsilon > 0$, there exists a natural number $N(\epsilon)$ such that for $n > N$, $|f_n - f| < \epsilon, \forall x \in D$. Preserves continuity.

Lebesgue integral: Partition the range of the function instead of the domain. For $0 \leq f_{\text{min}} \leq f(x) \leq f_{\text{max}}$, the range is partitioned into $f_{\text{min}} = f_1, f_2, \ldots, f_n = f_{\text{max}}$. For each f_k, there are sets of values X_k such that $f_k \leq f(x) < f_{k+1}$ for $x \in X_k$, and the size (measure) of each set is $\mu(X_k)$.

The Lebesgue integral is the limiting value of the sum:

$$\int f d\mu = \lim_{n\to\infty} \sum_{k=1}^{n} f_k \mu(X_k) = \lim_{n\to\infty} \sum_{\max|f_k - f_{k-1}| \to 0}^{n} f_k \mu(X_k)$$

Advantages:

- Only requires a measure of sets on the domain of integration – more general
- Change of order of limit and integration only requires pointwise convergence.
- A Riemann integrable funcion is also Lebesgue integrable, but not vice versa.

Measure of sets: A set X and a set Σ generated from X (contains X, $\emptyset$, and other subsets of X) defines a *measurable set* $\{X, \Sigma\}$. Along with a measure μ it defines a *measure space* $\{X, \Sigma, \mu\}$.

Rules for measure:

- If $\mu(E) \geq 0$ for all $E \in \Sigma$, all elements in Σ are measurable.

- $\mu(\emptyset) = 0$.
- If $E_i \subseteq E_j$, then $\mu(E_i) \leq \mu(E_j)$.
- $\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i)$ for disjoint sets $E_i \in \Sigma$.
- If (X, Σ_X) and (Y, Σ_Y) are both measureable, then a function $f : X \to Y$ is measurable if $f^{-1}(B) \in \Sigma_X$ for all $B \in \Sigma_Y$.

Lebesgue measure:

- For a continuous line segment on $\mathbb{R}$, the Lebesguemeasure is the length of the segment. e.g. $I = [a, b]$, $\mu(I) = b - a$.
- The Lebesgue measure of a single point is zero. By additivity, the Lebesgue measure for the set of rational numbers $\mu(\mathbb{Q}) = 0$.
- The Lebesgue meausre of countable sets (any set that can be put into a one-to-one correspondence with the set of integers $\mathbb{Z}$) is zero.

Probability measures: The probability space $\{\Omega, \mathcal{F}, P\}$ consists of three elements:

- Sample space Ω: the set of all possible outcomes.
- σ-algebra $\mathcal{F}$: certain subsets of Ω with outcomes to be considered.
- Probability measure P: a function that assigns a probability to each event in $\mathcal{F}$, satisfying $P(\Omega) = 1$ and countable additivity.

Random variables: For a probability space $\{\Omega, \mathcal{F}, P\}$ and a measurable space $\{E, \mathcal{E}\}$, there is an associated measurable function $X : \Omega \to E$ that satisfies $X^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{E}$. If $E = \mathbb{R}$, then $X : \Omega \to \mathbb{R}$ is a real-valued *random variable* such that $\{\omega \in \Omega : X(\omega) \leq r\} \in \mathcal{F}$ for every $r \in \mathbb{R}$.

The expectation of a random variable X is defined as $\mathbb{E}[X] = \int_\Omega X(\omega) dP(\omega)$.

Radon-Nikodym theorem: Consider two measures μ and ν on a measurable space $\{X, \Sigma\}$. If for every $A \in \Sigma$ where $\nu(A) = 0$ also have $\mu(A) = 0$, then μ is absolutely continuous with respect to ν, denoted as $\mu \ll \nu$.

Given a probability mesure P, a new probability measure Q can be defined if there exists a function $L(\omega)$ such that $dQ(\omega) = L(\omega) dP(\omega)$.

For any measurable set B, $Q(B) = \int_B L(\omega) dP(\omega)$ is the probability of B under the new measure Q.

The expectation of a function $F(\omega)$ can be computed as

$$\int_\Omega F(\omega) dQ(\omega) = \int_\Omega F(\omega) L(\omega) dP(\omega)$$

where the function $L(\omega) = \frac{dQ(\omega)}{dP(\omega)}$ is the Radon-Nikodym derivative, subject to (1) $L(\omega) \geq 0$ for all $\omega \in \Omega$ and (2) $\int_\Omega L(\omega) dP(\omega) = 1$.

For a random variable X on $\mathbb{R}$, the probability of a small neighborhood B around the outcome $X = x \in \mathbb{R}$ is $\int_B dP = \int_B L(x) d\mu(x) = \int_B L(x) dx$. The function $L(x) = \frac{dP(x)}{dx}$ is the probability density function of P with respect to the Lebesgue measure.

3 Markov Chain Monte Carlo

Probability inversion: For data $\mathrm{x} \in \mathcal{X}$ and parameters $\boldsymbol{\theta} \in \Theta$, the posterior distribution is $\pi(\boldsymbol{\theta}|\mathrm{x}) = \frac{p(\mathrm{x}|\boldsymbol{\theta})\pi_0(\boldsymbol{\theta})}{\int_\Theta p(\mathrm{x}|\boldsymbol{\theta})\pi_0(\boldsymbol{\theta})d\boldsymbol{\theta}}$. Inference requires integration with respect to the posterior

$$I(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{\theta}|\mathrm{x}}[f(\boldsymbol{\theta})] = \int_\Theta f(\boldsymbol{\theta})\pi(\boldsymbol{\theta}|\mathrm{x})d\boldsymbol{\theta}$$

The Monte-Carlo estimate is

$$\hat{I}_N(\boldsymbol{\theta}) = \frac{1}{N}\sum_{n=1}^{N} f(\boldsymbol{\theta}_n) \text{ with } \boldsymbol{\theta}_n \sim \pi(\boldsymbol{\theta}|\mathrm{x})$$

The estimator is unbiased and consistent as $N \to \infty$, $\hat{I}_N(\boldsymbol{\theta}) \to I(\boldsymbol{\theta})$ and the error $\sqrt{N}(\hat{I}_N(\boldsymbol{\theta}) - I(\boldsymbol{\theta})) \to \mathcal{N}(0, \sigma_f^2)$.

Markov chains: stochastic transitions governed by the Markov property

$$p(\boldsymbol{\theta}^{(i)}|\boldsymbol{\theta}^{(i-1)}, \ldots, \boldsymbol{\theta}^{(1)}) = T(\boldsymbol{\theta}^{(i)}|\boldsymbol{\theta}^{(i-1)})$$

Invariant distribution: the Markov chain converges to the invariant distribution $\pi(\boldsymbol{\theta}^{(\infty)}) \to \pi(\boldsymbol{\theta})$ irrespective of the initial state.

Irreducible: the Markov chain can reach any state from any other state in a finite number of steps.

Aperiodic: the Markov chain does not get trapped in cycles.

Detailed balance (reversible): a Markov chain satisfies detailed balance if $\pi(\boldsymbol{\theta}^{(i)})T(\boldsymbol{\theta}^{(i-1)}|\boldsymbol{\theta}^{(i)}) = \pi(\boldsymbol{\theta}^{(i-1)})T(\boldsymbol{\theta}^{(i)}|\boldsymbol{\theta}^{(i-1)})$.

3.1 Metropolis-Hastings Algorithm

Define the *transition kernel* as

$$\underbrace{P(x, dy)}_{\text{prob. of transition from } x \text{ to } dy} = \underbrace{p(x, y)dy}_{\text{prob. of moving to } dy \text{ around } y} + \underbrace{r(x)\delta_x(dy)}_{\text{prob. of staying at } x}$$

where $r(x) = 1 - \int p(x, y)dy$ is the probability of staying at x.

If $p(x, y)$ satisfies detailed balance $\pi(x)p(x, y) = \pi(y)p(y, x)$, then $\pi(\cdot)$ is the invariant density of the transition kernel $P(x, \cdot)$. In this case, the chain is *reversible*, satisfiying

$$P(y, dx)\pi(dy) = P(x, dy)\pi(dx)$$

To achieve this, the *proposal density* $q(x, y)$ is balanced with an *acceptance probability* that satisfies $\pi(x)q(x, y)\alpha(x, y) = \pi(y)q(y, x)\alpha(y, x)$.

The Metropolis-Hastings algorithm have $p_{\mathrm{MH}}(x, y) = q(x, y)\alpha(x, y)$ for $x \neq y$ with the acceptance probability

$$\alpha(x, y) = \min\left(1, \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)}\right)$$

Overall: Simulate y from the proposal density $q(x^{(i)}, y)$ and accept the move with probability $\alpha(x^{(i)}, y)$. If the proposed move is accepted, the next state is $x^{(i+1)} = y$. If the proposed move is rejected, the current value is repeated as the next state, i.e. $x^{(i+1)} = x^{(i)}$.

For symmetric proposal density $q(x, y) = q(y, x)$, the acceptance probability simplifies to $\alpha(x, y) = \min\left(1, \frac{\pi(y)}{\pi(x)}\right)$: move to higher density regions are always accepted.

3.2 Gibbs Sampling

Product transition kernels: the product kernel $P_1(x_1, dy_1, |x_2)P_2(x_2, dy_2|y_1)$ has invariant density $\pi(y_1, y_2)$ if:

- the conditional kernel $P_1(x_1, dy_1|x_2)$ has invariant density $\pi_{1|2}(\cdot|x_2)$ for fixed x_2
- the conditional kernel $P_2(x_2, dy_2|x_1)$ has invariant density $\pi_{2|1}(\cdot|x_1)$ for fixed x_1.

Gibbs sampler: set $P_1(x_1, dy_1|x_2) = \pi^*_{1|2}(dy_1|x_2)$ and $P_2(x_2, dy_2|x_1) = \pi^*_{2|1}(dy_2|x_1)$. For the move of x_1 with x_2 fixed, use exact conditional densities as the proposal denisites, i.e. $q([x_1, x_2], [y_1, y_2]) = \pi(y_1|x_2)$ and $q([y_1, y_2], [x_1, x_2]) = \pi(x_1|y_2)$. As a result, the acceptance probability is $\alpha = 1$ for all moves.

Limitation: small step sizes and poor mixing if the components are highly correlated.

Mixture transition kernels: mixing transitional kernels that have the same invariant density, i.e. $P(x, dy) = \sum_i w_i P_i(x, dy)$ with $\sum_i w_i = 1$.

Advantage: for multimodal distributions, the mixture kernel can have better mixing properties by combining local and global moves.

3.3 Data Augmentation

Augment a target density $\pi(\theta)$ with ϕ to form $\pi(\theta, \phi)$. Draw samples $\theta^{(n)}, \phi^{(n)} \sim \pi(\theta, \phi)$, then for each $\theta^{(n)}$ it follows $\int \pi(\theta^{(n)}, \phi)d\phi = \pi(\theta^{(n)})$.

4 Vector, Metric and Banach Spaces

Vector space V: a collection of vectors x with axioms (group):

- $x + y = y + x$ (commutativity)
- $(x + y) + z = x + (y + z)$ (associativity)
- There exists a zero vector such that $x + 0 = x$ for all x
- For each x, there exists an additive inverse such that $x + (-x) = 0$
- For $\alpha \in \mathbb{F}$, $\alpha x \in V$ for all $x \in V$ (scalar multiplication is closed)
- $\alpha(x + y) = \alpha x + \alpha y$ (distributivity)
- $\alpha(\beta x) = (\alpha\beta)x$ (associativity of scalar multiplication)
- $0 \times x = 0, 1 \times x = x$

Inner product space: space with an inner product $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ satisfying:

- $\langle x, y\rangle = \overline{\langle y, x\rangle}$
- $\langle \alpha x + \beta y, z\rangle = \alpha\langle x, z\rangle + \beta\langle y, z\rangle$ for $\alpha, \beta \in \mathbb{R}$
- $\langle x, x\rangle \geq 0$ and $\langle x, x\rangle = 0$ if and only if $x = 0$
- For $a, b \in \mathbb{C}^n$, $\langle a, b\rangle = \sum_{i=1}^{n} a_i^* b_i$

Inner product of functions: $\langle f, g\rangle = \int_a^b f(x)g(x)dx$ for $x \in [a, b] \in \mathbb{R}$.

Vector norm: the length of a vector, defined as $\|x\| = \sqrt{\langle x, x\rangle}$. The norm satisfies:

- Schwarz inequality: $|\langle x, y\rangle| \leq \|x\|\|y\|$
- Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$
- Parallelogram law: $\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$

Orthogonality: two elements x and y are orthogonal if $\langle x, y\rangle = 0$. e.g. $\sin(x)$ and $\cos(x)$ are orthogonal on $[-\pi, \pi]$.

Orthonormal basis: a finite set of linearly independent, orthogonal unit vectors with which every vector can be expressed as $x = \sum_{i=1}^{n} \alpha_i e_i$.

Cauchy sequence: a sequence of vectors $\{x_n\}$ is a Cauchy sequence if for any $\epsilon > 0$, there exists a natural number N such that for all $m, n > N$, $\|x_n - x_m\| < \epsilon$.

If every Cauchy sequence in a vector space converges to a limit that is also in the vector space, then the vector space is *complete*.

Metric vector space: a vector space with a distance function $\rho(x, y)$ that maps two vectors to a non-negative real number, satisfying:

- $\rho(x, y) = 0$ if and only if $x = y$
- $\rho(x, y) = \rho(y, x)$ (symmetry)
- $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$ (triangle inequality)

Normed space: a metric space with distance function $\rho(x, y) = \|x - y\|$ induced by a non-negative norm $\|\cdot\|$.

Banach space: a complete normed space. All finite-dimensional normed spaces are Banach spaces.

The l_p space is a Banach space where $\sum_{i=1}^{\infty} |x_i|^p < \infty$ for $p \geq 1$, and the norm is the l^p norm $\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}$.

The L_p space is a Banach space where $\int_a^b |f(x)|^p dx < \infty$, where x is the Lebesgue measure. The norm is the L^p norm $\|f\|_p = \left(\int_a^b |f(x)|^p dx\right)^{1/p}$.

5 Hilbert Spaces

Pre-Hilbert space: an incomplete inner product space.

Hilbert space: a complete inner product space. For any f in a Hilbert space with an infinite set of orthonormal vectors $\{e_i\}$,

$$\{e_i\} \text{ is complete} \iff \|f\|^2 = \sum_{i=1}^{\infty} |\langle f, e_i \rangle|^2 \quad \text{(Parseval's identity)}$$

l^2 and L^2 spaces are both Hilbert spaces.

Equivalence of l^2 and L^2 spaces: a function $f \in L^2$ can be represented with a sequence $\{c_k = \langle f, \phi_k \rangle\} \in l^2$ by Fourier series

$$f = \sum_{k=1}^{\infty} c_k \phi_k$$

where $\{\phi_k\}$ is an orthonormal basis of L^2.

l^2 and L^2 are isomorphic, i.e. there is a one-to-one correspondence between the elements of l^2 and L^2. l^2 is a coordinate system for L^2.

Reproducing kernel Hilbert space (RKHS): a Hilbert space of functions where point evaluation is continuous i.e. functions close in norm are also close pointwise.

The *reproducing kernel function* $K_x = K(x, y)$ reproduces function evaluation by inner product $f(x) = \langle f, K_x \rangle$. The kernel function is symmetric and positive definite, and uniquely defines the RKHS.

Function approximation in RKHS: Given data $\{(x_i, y_i)\}_{i=1}^{n}$ where $y_i = f(x_i)$, the regularised empicical error

$$\frac{1}{N} \sum_{n=1}^{N} |y_i - f(x_i)|^2 + \lambda \|f\|^2$$

is minimised by functions of the form

$$\hat{f}(\cdot) = \sum_{n=1}^{N} \alpha_n K(x_n, \cdot)$$

where $\alpha = (K + N\lambda I)^{-1} y$ and K is the kernel matrix with $K_{ij} = K(x_i, x_j)$. The term $\lambda \|f\|^2$ penalises the complexity of the function and prevents overfitting.

6 Sobolev Spaces

Sobolev space: a Sobolev space $W_p^k(\Omega)$ is a Banach space where functions $f \in L^p(\Omega)$ have *weak derivatives* $D^s f \in L^p(\Omega)$ for $0 \leq s \leq k$. The norm is defined as

$$\|f\|_{W_p^k} = \left(\sum_{s=0}^{k} \int_{\Omega} |D^s f(x)|^p \, dx \right)^{1/p}$$

For $p = 2$, the Sobolev space $W_2^k(\Omega)$ is a Hilbert space with inner product

$$\langle f, g \rangle_{W_2^k} = \sum_{s=0}^{k} \langle D^s f, D^s g \rangle = \sum_{s=0}^{k} \int_{\Omega} D^s f(x) D^s g(x) \, dx$$

The norm and the inner product in $W_2^k(\Omega)$ encodes the smoothness of function.

Weak derivatives: for functions $f : [0,1] \to \mathbb{R}$, a function $h : [0,1] \to \mathbb{R}$ is a weak derivative of f if for every differentiable function $g : [0,1] \to \mathbb{R}$ with $g(0) = g(1) = 0$,

$$\int_0^1 f(x)g'(x)dx = -\int_0^1 h(x)g(x)dx$$

Function approximation by polynomial expansion: By Fourier expansion, functions from the set $C_2[-\pi,\pi] = C[-\pi,\pi] \cap L^2[-\pi,\pi]$ can be presented as $f(x) = \sum_{k=0}^{\infty} c_k \exp(ikx)$ with $c_k = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\exp(-ikx)dx$. The L^2 norm of f is $\|f\|^2 = \sum_{k=0}^{\infty} c_k^2$.

The approximating functions $f_n(x) = \sum_{k=0}^{n} c_k \exp(ikx)$ are in the Sobolev space W_2^s with the norm $\|f\|_{W_2^s}^2 = \|D^s f\|_{L^2}^2 = \sum_{k=0}^{\infty} k^{2s} c_k^2$.

The appoximation error is bounded

$$\|f - f_n\|_{L^2}^2 = \sum_{k=n+1}^{\infty} c_k^2 < \frac{1}{n^{2s}} \|f\|_{W_2^s}^2$$

as s increases, the approximation error decreases faster with increasing n.

For d-dimensional domains, the error bound is $\|f - f_n\|_{L^2}^2 < \frac{1}{n^{2s/d}} \|f\|_{W_2^s}^2$, which shows the curse of dimensionality – convergence rate decreases as d increases.

7 Gaussian Measures in Hilbert Space

Lebesgue measure in D dimensions: $\mu = \prod_{d=1}^{D} \mu_d$.

There is no Lebesgue measure in infinite dimensional Hilbert or Banach spaces.

Gaussian measure in infinite dimensions: a Gaussian measure is a probability measure characterised by a mean function and a covariance operator.

Diagonal covariance operator: $S_x = (a_n x_n)_{n=1}^{\infty}$ for $x \in l^2$ subject to $\sum_{k=1}^{\infty} a_k < \infty$.

Full covariance operators: need to be trace class, i.e. $\mathrm{Tr}(C) = \sum_{n=1}^{\infty} \langle e_k, Ce_k \rangle < \infty$ for an orthonormal basis $\{e_k\}$. Given an covariance function $c(x,y)$

$$Ce_k(x) = \int c(x,y) e_k(y) dy$$

Gaussian measures in Hilbert space: In a Hilbert space, two Gaussian measures are either equivalent or singular.

Equivalent: also known as mutually absolutely continuous, two measures P and Q are equivalent if $P(A) = 0$ if and only if $Q(A) = 0$ for any measurable set A.

Singular: two measures P and Q are singular if there exists a measurable set A such that $P(A) = 1$ and $Q(A) = 0$.

For two Gaussian measures $\mathcal{N}(0, C)$ and $\mathcal{N}(v, C)$ in l^2, two measures are equivalent if $v \in \mathrm{Image}(C^{1/2})$ i.e. $v = C^{1/2}u$ for some $u \in l^2$, and singular otherwise. The subspace for v is the Cameron-Martin space, also the reproducing kernel Hilbert space (RKHS).

Bayes' rule in Hilbert space: The Radom-Nikodym derivative of the posterior probability measure with respect to the prior probability measure is given by

$$\frac{d\mu^y}{d\mu^0} \propto \underbrace{P(y|x)}_{\text{likelihood}}$$

In a infinite dimensional Hilbert space, the prior measure over functions is a Gaussian measure $\mu_0 = \mathcal{N}(m, C)$ – formal definition of Gaussian Process prior.

8 MCMC in Infinite-Dimensional Spaces

Metropolis-Hastings algorithm in infinite dimensions: Define a proposal

$$v = \sqrt{1-\beta^2}u + \beta\xi, \xi \sim \mathcal{N}(0, C)$$

such that u and v both follow $\mathcal{N}(0, C)$. Given a prior Gaussian measure $\mu^0 = \mathcal{N}(0, C)$ and a likelihood $P(y|v) = \exp(-\Phi(v))$, the acceptance probability (replacing $\pi(v)q(v, u)$ with $\mu^y(dv)q(u, dv)$, where μ^y is the posterior measure) is

$$\alpha(u, v) = \min\left(1, \frac{\mu^y(dv)q(v, du)}{\mu^y(du)q(u, dv)}\right) = \min\left(1, \frac{P(y|v)}{P(y|u)}\right) = \min\left(1, \exp(\Phi(u) - \Phi(v))\right)$$

9 Langevin Dynamics

Langevin stochastic differential equation (SDE):

$$dX_t = -\nabla U(X_t)dt + \sqrt{2}dB_t = \nabla \log \pi(X_t)dt + \sqrt{2}dB_t$$

where $U(\cdot) : \mathbb{R}^d \to \mathbb{R}$ is a potential function, B_t is the Brownian motion and $\pi(x) \propto \exp(-U(x))$ is the stationary distribution of the SDE.

The measure of X_t, ρ_t, satisfies the Fokker-Planck equation

$$\frac{\partial \rho_t}{\partial t} = \nabla \cdot (\rho_t \nabla \log \pi) + \Delta \rho_t$$

with asymptotic solution $\rho_\infty = \pi$.

Log-concave: A probability measure with density π is log-concave if

$$\pi(\lambda x + (1-\lambda)y) \geq \pi(x)^\lambda \pi(y)^{1-\lambda} \text{ for } 0 < \lambda < 1$$

A probability measure with density π is m-strongly log-concave if

$$\pi(\lambda x + (1-\lambda)y) \geq \pi(x)^\lambda \pi(y)^{1-\lambda} \exp\left(\frac{m}{2}\lambda(1-\lambda)\|x-y\|^2\right) \text{ for } 0 < \lambda < 1$$

i.e. there are no flat regions and the maximum is unique.

Convergence of Langevin dynamics: For an m-strongly log-concave π, the convergence rate to the stationary measure is exponentially fast

$$W_2(\rho_t, \pi) \leq e^{-mt}\left\{\|x - x^*\| + \left(\frac{d}{m}\right)^{\frac{1}{2}}\right\}$$

for $\pi_0 = \delta_x$ and $x^* = \arg\max_x \pi(x)$.

Metropolis-adjusted Langevin algorithm (MALA): Given a state of the Markov chain X_k, propose a new state $\bar{X}_{k+1}$ by first-order Euler discretisation with step size γ:

$$\bar{X}_{k+1} = X_k + \gamma \nabla \log \pi(X_k) + \sqrt{2\gamma} W_k, W_k \sim \mathcal{N}(0, I)$$

To correct the discretisation bias, accept the proposed move with probability

$$\alpha(X_k, \bar{X}_{k+1}) = \min\left(1, \frac{\pi(\bar{X}_{k+1}) q(X_k|\bar{X}_{k+1})}{\pi(X_k) q(\bar{X}_{k+1}|X_k)}\right)$$
$$\text{where } q(\bar{X}_{k+1}|X_k) \propto \exp\left(-\frac{1}{4\gamma}\|\bar{X}_{k+1} - X_k - \gamma\nabla \log \pi(X_k)\|^2\right)$$

Limitation: the acceptance probability is exponentially low for high dimension, leading to poor mixing.

Unadjusted Langevin algorithm (ULA): Run the first-order Euler discretisation without the Metropolis-Hastings correction, i.e. $X_{k+1} = X_k + \gamma \nabla \log \pi(X_k) + \sqrt{2\gamma} W_k$. The stationary distribution is a biased distribution π_γ, with asymptotic bias of order $\mathcal{O}(\gamma^{\frac{1}{2}})$.

Langevin schemes for Bayesian inference: For a target distribution $\pi(x) \propto p(x) \prod_{i=1}^n p(y_i|x)$, where $p(x)$ is the prior and $p(y_i|x)$ is the likelihood for data vector y_i, samples can be generated from

$$X_{k+1} = X_k + \gamma\left(\nabla \log p(X_k) + \sum_{i=1}^n \nabla \log p(y_i|X_k)\right) + \sqrt{2\gamma} W_k$$

For MALA, the complexity is $\mathcal{O}(n)$ for both the gradient and the acceptance probability (can be avoided in ULA) computation.

Stochastic gradient Langevin dynamics (SGLD): An unbiased estimate of the gradient can be computed more efficiently by subsampling the data. At each iteration k, choose a random index set $\mathcal{I}_k \subset \{1, \ldots, n\}$ and approximate the gradient

$$\widehat{\nabla \log \pi(X_k)} = \nabla \log p(X_k) + \frac{n}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \nabla \log p(y_i|X_k)$$

The complexity is $\mathcal{O}(|\mathcal{I}_k|)$, and convergence rate is similar to ULA.

10 Energy-Based and Score-Based Models

Generative models: approximate the data distribution p_{data} with a model distribution p_θ, by learning parameters θ from data $\mathcal{D} = \{x_i\}_{i=1}^N$ drawn from p_{data}.

Energy-based models (EBMs): The probability of a state x is defined as

$$p(x) = \frac{1}{Z_\beta} \exp(-\beta E(x))$$

where $E(x)$ is the energy function, β is the inverse temperature and $Z_\beta = \int \exp(-\beta E(x))dx$ is the partition function. EBMs paratermises the energy function, giving $p_\theta(x) \propto \exp(-E_\theta(x))$.

The EBMs are trained via maximum likelihood or contrastive divergence.

Maximum likelihood: maximise the log-likelihood $\mathcal{L}(\theta) = \mathbb{E}_{x\sim p_{\text{data}}}[\log p_\theta(x)]$ via gradient methods, with gradient:

$$\nabla_\theta \mathcal{L}(\theta) = -\mathbb{E}_{x\sim p_{\text{data}}}[-\nabla_\theta E_\theta(x)] + \underbrace{\mathbb{E}_{x\sim p_\theta}[-\nabla_\theta E_\theta(x)]}_{\nabla_\theta \log Z_\theta}$$

Limitation: the second term requires sampling from the model distribution p_θ, which is computationally expensive.

Contrastive divergence:

1. For each x_i, initialise a MCMC chain with target distribution p_θ at $x_i^{(0)}$.
2. Run the MCMC chain for k steps to obtain $x_i^{(k)}$.
3. Compute the gradient estimate

$$\hat{\mathcal{L}} = -\frac{1}{N}\sum_{i=1}^{N}\left(-\nabla_\theta E_\theta(x_i^{(0)}) + \nabla_\theta E_\theta(x_i^{(k)})\right)$$

4. Update parameters $\theta \leftarrow \theta + \eta\hat{\mathcal{L}}$.

Limitations: bias in the gradient estimate and MCMC can be very slow.

Score-based models: Learn to match the score function $s_\theta(x) = \nabla_x \log p_\theta(x) = -\nabla_x E_\theta(x)$, bypassing the partition function. Trained by score matching, which minimises the Fisher divergence between p_{data} and p_θ:

$$J(\theta) = \frac{1}{2}\mathbb{E}_{x\sim p_{\text{data}}}\left[\|s_\theta(x) - \underbrace{\nabla \log p_{\text{data}}(x)}_{\text{intractable}}\|^2\right]$$

which can be rewritten as

$$J_{\text{SM}}(\theta) = \mathbb{E}_{x\sim p_{\text{data}}}\left[\underbrace{\text{Tr}(\nabla_x s_\theta(x))}_{\text{expensive for high-dim.}} + \frac{1}{2}\|s_\theta(x)\|^2\right]$$

Sliced score matching: approximate the trace of the Jacobian by projecting the score function onto random directions $v \sim \mathcal{N}(0, I)$

$$J_{\text{SSM}}(\theta) = \mathbb{E}_{x\sim p_{\text{data}}, v\sim\mathcal{N}(0,I)}\left[v^\top \nabla_x s_\theta(x) v + \frac{1}{2}\|s_\theta(x)\|^2\right]$$

In practice, one sample of v is used for each data point:

$$\hat{J}_{\text{SSM}}(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left[v_i^\top \nabla_x s_\theta(x_i) v_i + \frac{1}{2}\|s_\theta(x_i)\|^2\right]$$

Sampling from generative models: use Langevin dynamics

$$dx_t = s_\theta(x)dt + \sqrt{2}dB_t$$

which have stationary distribution p_θ. In practice, discretise the SDE with step size γ:

$$x^{(k+1)} = x^{(k)} + \gamma s_\theta(x^{(k)}) + \sqrt{2\gamma}z^{(k)}, \quad z^{(k)} \sim \mathcal{N}(0, I)$$

Parallel tempering (replica exchange): For $\pi(x) = \exp(-U(x))$, define the tempered (smoothed) distribution $\pi_\beta(x) = \exp(-\beta U(x))$ with inverse temperature β. Run MCMC chains on multiple tempered distributions $\{\pi_{\beta_i}\}$ in parallel, and exchange samples between i-th and j-th chains with acceptance probability:

$$\alpha = \min\left(1, \frac{\pi_{\beta_i}(x_t^j)\pi_{\beta_j}(x_t^i)}{\pi_{\beta_i}(x_t^i)\pi_{\beta_j}(x_t^j)}\right)$$